

Homework 2

Handed Out: Tue September 29, 8:00 p.m. ET

Due: Tue October 20, 7:59 p.m. ET

- Submit written HW2 as one legible PDF through Gradescope. You may write your answers in $\text{\LaTeX}$ or scan handwritten work. We cannot award credit for work that cannot be read.
- Assign the correct PDF pages to every question and sub-question in Gradescope. Unassigned pages may receive no credit.
- The submission deadline is **7:59 p.m. ET**. The late penalty is 0.5% of the assignment's maximum score per late hour (rounded up), for at most 48 late hours. Written and coding submissions are timed separately.
- The written and coding halves are graded separately. Written HW2 contains 55 points for CIS 5190 students. The 5-point part marked [**CIS 5190 only**] is optional for CIS 4190 students, whose written maximum is therefore 50 points. Optional work cannot raise a score above 100%.
- Follow the [Human and AI Assistance in Homework policy](#). Do not share or discuss solutions with other students. AI use is permitted only after a documented solo attempt and must be disclosed below.
- If using $\text{\LaTeX}$, copy the [shared solutions template](#). The link is read-only; make a *private* copy before editing.
- Post clarification questions on Ed Discussion. Contact the course staff promptly if you experience submission problems.

1 Use of AI Tools

Report any use of AI tools (ChatGPT, Claude, Copilot, Gemini, Cursor, etc.) while working on **this written homework**. You report AI use for the coding homework separately, in the reporting cell at the end of the notebook.

Being honest here will not cost you points. This section is graded for being answered, not for any particular answer: reporting that you used no AI earns full marks, and so does reporting that you used a lot of it. We are trying to learn what actually helps people learn, which only works if the reports are accurate.

1. (2 pts) Did you use AI tools on this written homework? If **no**, say so and you are done with this section. If **yes**, also answer:
 - Which tools did you use?

- **How** did you use them? Be specific: checking an answer you had already derived, explaining a concept you were stuck on, generating a first draft you then rewrote, doing the derivation for you, etc.
- How do you think this affected your learning of this material? Say so if you think it helped, and equally if you think it let you skip understanding something you would otherwise have worked out.

2 Time Spent

1. (2 pts) Roughly how long did each part of this **written** homework take, in hours? Decimals are fine ($0.5 = 30$ minutes), and enter 0 for any question you skipped. Report a time for each of Questions 1–6 as well as for reading the assignment. Rough estimates are genuinely useful — please do not agonize over precision, but do fill this in **as you go** rather than reconstructing it at the end. Also graded for being answered, not for what you answer.

3 Written Questions

You do not need to show work for multiple-choice items unless a question says otherwise. For derivations and calculation questions, show enough work for the grader to follow your reasoning; an unsupported final answer may not receive full credit.

1. [Logistic Regression] (8 pts)

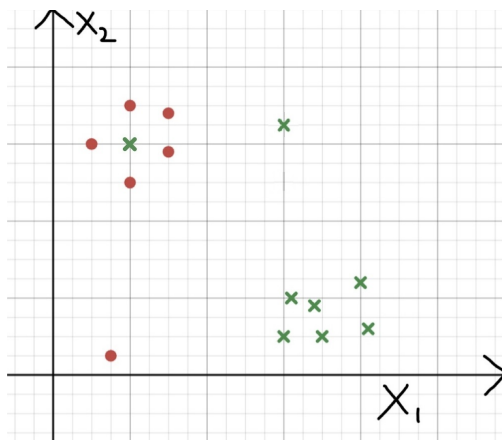


Figure 1: Data for Logistic Regression Question

Let the data distribution, as shown in Figure 1, represent the binary classification problem where we fit the model $p(y = 1|x, \beta) = \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2)$. As seen in class, we do this by minimizing the negative log-likelihood (log loss), which is equivalent to maximizing the likelihood:

$$L(\beta) = -\ell_Z(\beta)$$

where $\ell_Z(\beta)$ represents the log-likelihood on the training set Z . In the figure, the red circles have label $y = 1$ and the green crosses have label $y = 0$.

For the questions below, submit the answer to each question as a separate figure. We just expect an approximation of the figures if you submit hand-drawn solutions, also be careful about the clarity of your submitted figures. For parts (b)–(d), assume that λ is effectively infinite and that only the named coefficient is regularized. Treat that coefficient as 0.

- (a) (2 pts) Show a decision boundary that could correspond to the fitted parameters $\hat{\beta}$ after training the classifier. How many datapoints are wrongly classified on the training data?
- (b) (2 pts) For this part, strong regularization is applied only to the β_0 parameter, and we minimize

$$L_0(\beta) = -\ell_Z(\beta) + \lambda\beta_0^2$$

Draw a decision boundary that approximately corresponds to the optimal classifier with parameters $\hat{\beta}$. How many datapoints are wrongly classified on the training data?

- (c) (2 pts) Now, the same strong regularization is performed only on the β_1 parameter, i.e., we minimize

$$L_1(\beta) = -\ell_Z(\beta) + \lambda\beta_1^2$$

Show a decision boundary that possibly would correspond to $\hat{\beta}$. How many datapoints are wrongly classified on the training data?

- (d) (2 pts) Finally, the same strong regularization is applied only to the β_2 parameter.

$$L_2(\beta) = -\ell_Z(\beta) + \lambda\beta_2^2$$

Show a decision boundary that possibly would correspond to $\hat{\beta}$. How many datapoints are wrongly classified on the training data?

2. [Neural Networks] (6 pts)

- a. (2 pts) Consider a feedforward neural network with at least one hidden layer. Suppose all weights and biases are initialized to zero. What problem does this cause during training?
- b. (2 pts) Why do we use a non-linear activation function in a feed forward neural network?
- c. (2 pts) Consider a fully connected neural network with the following architecture:
Input layer: 40×30 grayscale image
Hidden layer 1: 64 units
Hidden layer 2: 32 units
Output layer: 10 units

Assume biases are included in the network. Calculate the total number of trainable parameters in this network.

3. [**k Nearest Neighbors**] (6 pts)

Consider properties of KNN models:

- a. (2 pts) Suppose we are using 1-nearest neighbor (kNN with $k = 1$) for binary classification under Euclidean distance.
 - i. If the training set contains only two points with different labels, describe the resulting decision boundary. [Hint: Try drawing a picture]
 - ii. Now assume we have full control over the training dataset and can make it arbitrarily large. Can a 1-NN classifier approximate any desired decision boundary as closely as we want? If yes, explain how the training points should be arranged. If no, give an example of a boundary it cannot approximate.
- b. (2 pts) Suppose we take $k = n$, where n is the size of the finite training dataset. What would kNN output for a regression problem? What about for a classification problem?
- c. (2 pts) What effect does increasing the number of nearest neighbors k have on the bias-variance tradeoff? Explain your answer. [Hint: Use parts (a) and (b) in your explanation.]

4. [**k Nearest Neighbors**] (6 pts) Imagine you want to apply k Nearest Neighbors to the binary classification problem of predicting whether a house is worth more than \$600, 000 ($y = +1$) or less than \$600, 000 ($y = -1$). Suppose you have two features: the square footage of the house, and the distance of the house from the nearest school (in miles). Most houses in the dataset are located in residential neighborhoods with schools nearby.

- a. (2 pts) Why might Euclidean distance between feature vectors representing houses be a bad metric of similarity / dissimilarity? What might you do to solve this problem?
- b. (2 pts) Suppose you found another potential feature: the type of roof the house has, expressed as a categorical feature via integer IDs (i.e. 0 = shingles, 1 = padded, etc.). Is it a good idea to use this feature alongside the other two features, with Euclidean distance? Explain why or why not. If not, what encoding would be more appropriate?
- c. (2pts) Briefly describe the curse of dimensionality. Why can k -NN still perform well on some datasets like the handwritten digits, despite these images being very high dimensional?

5. [**Decision Trees**] (8 pts)

In class, we discussed early stopping of generating splits and post-pruning. Here, we consider how they interact.

- a. (2 pts) Decision tree training often combines early stopping criteria (such as maximum depth or minimum samples per split) with post-pruning on a validation set. If post-pruning can remove unnecessary branches, why might it still be useful to apply early stopping during tree construction?
 - b. (4 pts) Suppose we are training a decision tree with both early-stopping conditions and post-pruning. For each of the following, describe how it affects bias and variance: (i) increase the maximum depth of the decision tree, (ii) increase the minimum number of samples needed to split, (iii) disable post-pruning, and (iv) assuming we are using a feature map, add more features to the feature map.
 - c. (2 pts) For the given boolean functions, please provide the corresponding decision tree representations. Here, $\wedge$ denotes logical *AND* and $\vee$ denotes logical *OR*.
 - i. $A \wedge (B \vee C)$
 - ii. $(A \wedge B) \vee (C \wedge D)$
6. **[Decision Trees] (17 pts)** Consider the following set of training examples for a decision tree classifier. The attributes a_1 and a_2 take values $+$ or $-$.

Instance	a_1	a_2	Classification
1	-	+	F
2	-	-	T
3	+	+	T
4	+	+	T

Recall the following definitions of entropy (H), Information Gain, and Gini index, which are useful for this problem:

$$H(\mathcal{D}) = - \sum_c P(Y = c) \log_2 P(Y = c),$$

$$\text{IG}(\mathcal{D}, X_j) = H(\mathcal{D}) - \sum_v P(X_j = v) H(\mathcal{D}[X_j = v]),$$

$$\text{Gini}(\mathcal{D}) = \sum_c P(Y = c)(1 - P(Y = c)) = 1 - \sum_c P(Y = c)^2.$$

- a. (2 pts) What is the class entropy of this collection of training examples? Please write your final answer in decimal form.
- b. (6 pts) What is the information gain of the two attributes respectively relative to these training examples? Please write your final answer in decimal form.
- c. (3 pts) Draw the complete decision tree, showing the class predictions at the leaves.

Assuming you are using LaTeX, you may (i) very neatly hand draw the tree, photograph it, and include it as a figure, (ii) draw it using a graphics program or PowerPoint, (iii) express the tree in a series of if statements, preferably using LaTeX's `verbatim` environment, or (iv) typeset the tree directly with [TikZ](#).

- d. (1 pts) Given this decision tree, what would be the classification for a new instance with $a_1 = +$ and $a_2 = -$?
- e. (5 pts) **[5190 ONLY]** What is the Gini gain of each of the two attributes respectively relative to these training examples? Be sure to clearly show your work, including for Gini index of parent and weighted average Gini index of children.